

Cor lemm Schwarz derivada distancia borde imagen

Corolario 1. Sea $\Omega \subset \mathbb{C}$ un dominio y $f \in \mathcal{H}(\Omega) : \exists a \in \Omega, M > 0 : f(\Omega) \subset D(f(a), M)$

$$\implies |f'(a)| \leq \frac{M}{\text{dist}(a, \partial\Omega)}.$$

Demostración: Sea $R = \text{dist}(a, \partial\Omega)$. Consideramos la función $g: \mathbb{D} \rightarrow \mathbb{C}$ dada por

$$\forall z \in \mathbb{D} : g(z) = \frac{f(a + Rz) - f(a)}{M}.$$

Entonces, g es holomorfa en $\mathbb{D}$, $g(0) = 0$ y $\forall z \in \mathbb{D} : |g(z)| \leq 1$. Por el lema de Schwarz, $|g'(0)| \leq 1$. Por tanto, $|f'(a)| = \frac{M}{R} |g'(0)| \leq \frac{M}{\text{dist}(a, \partial\Omega)}$. ■

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